



## Jireh Magsino Tregueros

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### Education

#### Laguna State Polytechnic University - Los Baños Campus

Bachelor of Science in Computer Science; Specialization in Intelligent Systems

Los Baños, Laguna

2022-2026

#### Tuntungin-Putho Integrated National High School

Senior High School - General Academic Strand (GAS)

Junior High School

Los Baños, Laguna

Graduated 2022

Graduated 2020

### Projects

#### Microplastic Density Mapping in Southern Laguna de Bay

*Co-Developer – LSPU Los Baños, June 2026*

- Built hybrid MLP autoencoder + ensemble ML (RF, GB, XGBoost) model predicting microplastic concentrations from Sentinel-2 indices and in-situ water quality data.

#### IRIS AI: Intelligent AI Assistant

*Co-Developer - Web Application - 2024*

- Co-developed a responsive AI-powered web assistant with real-time conversational capabilities, modern user interface, and seamless AI integration.

#### ASTRA AI: Intelligent Multi-Modal Chat Assistant

*Web Development Project – 2025*

- Multi-modal chatbot with voice, themes, Markdown support; responsive design and real-time AI API integration.
- Deployed production-ready system with responsive design, accessibility features, and real-time interaction capabilities.

### Experience

#### Freelance Developer and Data Analyst

*Remote — May 2022 – Jun 2025*

- Designed AI and data-driven systems for environmental and educational applications.
- Performed data analysis, feature engineering, and model evaluation using Python and ML frameworks.
- Developed automation and web systems.

#### Database Manager (Internship)

*On-site — Laguna State Polytechnic University - Los Baños — Feb 2026 – Jun 2026*

- Managed and maintained institutional databases ensuring data accuracy and organization.
- Performed data cleaning, validation, and updates to support reporting and operations.

#### Programming Languages:

Python, Java, JavaScript (ES6+), HTML5, CSS3, SQL

#### Tech Stack:

React.js, JavaScript, HTML5, CSS3, Responsive Web Design, Markdown, Node.js, REST APIs, AI API Integration  
TensorFlow (Keras), scikit-learn, XGBoost, Random Forest, Gradient Boosting, NumPy, Pandas, Predictive Modeling,  
Feature Engineering, Data Analysis, Natural Language Processing  
Sentinel-2, Sentinel Hub, Remote Sensing, Environmental Data Analysis, QGIS  
MySQL, SQL Database Management  
Git, GitHub, VS Code, Google Colab, Microsoft Excel, Google Sheets, Claude Code, Copilot  
Web Development, AI Chatbots, Web-Based Automation, Real-Time Applications